

CODE:

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\# Program to implement Naïve Bayes Algorithm
# and display Confusion Matrix with Accuracy
# Program to implement Naïve Bayes Algorithm
# using Breast Cancer Dataset

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, accuracy_score

# Load Dataset
data = load_breast_cancer()

X = data.data
y = data.target

# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.3,
    random_state=42
)

# Create Naïve Bayes model
classifier = GaussianNB()

# Train the model
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classifier.fit(X_train, y_train)

# Predict the test data
y_pred = classifier.predict(X_test)

# Print Actual and Predicted Values
print("Actual Values:")
print(y_test)

print("\nPredicted Values:")
print(y_pred)

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)

# Extract TN, FP, FN, TP
TN, FP, FN, TP = cm.ravel()

print("\nDetailed Explanation")
print("-----")
print("True Positives (TP) :", TP)
print("True Negatives (TN) :", TN)
print("False Positives (FP):", FP)
print("False Negatives (FN):", FN)
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# Accuracy

accuracy = accuracy_score(y_test, y_pred)

print("\nAccuracy = {:.2f}%".format(accuracy * 100))

# Additional Performance Measures

precision = TP / (TP + FP)

recall = TP / (TP + FN)

specificity = TN / (TN + FP)

print("Precision = {:.2f}%".format(precision * 100))

print("Recall = {:.2f}%".format(recall * 100))

print("Specificity= {:.2f}%".format(specificity * 100))

```

OUTPUT:

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> | Actual Values:
[1 0 0 1 1 0 0 0 1 1 1 0 1 0 1 0 1 1 1 0 0 1 0 1 1 1 1 1 1 0 1 1 1 1 1 0
 1 0 1 1 0 1 1 1 1 1 1 1 1 0 0 1 1 1 1 1 0 0 1 1 0 0 1 1 1 0 0 1 1 0 0 1 0
 1 1 1 0 1 1 0 1 0 0 0 0 0 0 1 1 1 1 1 1 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1 0
 1 1 0 1 0 1 1 1 0 1 1 1 0 1 0 0 1 1 0 0 0 1 1 1 0 1 1 1 0 1 0 1 0 1 0 0
 0 1 0 1 1 1 1 0 0 1 1 1 1 1 1 1 0 1 1 1 1 0 1]

Predicted Values:
[1 0 0 1 1 0 0 0 1 1 1 0 1 0 1 0 1 1 1 0 1 1 1 1 1 1 1 0 1 1 1 1 1 1 0
 1 0 1 1 0 1 1 1 1 1 1 1 1 0 0 1 1 1 1 1 0 0 1 1 0 0 1 1 1 0 0 1 1 0 0 1 0
 1 1 1 1 1 1 0 1 1 0 0 0 0 0 1 1 1 1 1 1 1 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1 0
 1 1 0 0 0 1 1 1 0 0 1 1 0 1 0 0 1 1 0 0 0 1 1 1 0 1 1 0 0 1 0 1 1 0 1 0 0
 1 1 1 1 1 1 1 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0]

Confusion Matrix:
[[ 57   6]
 [  4 104]]

Detailed Explanation
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True Positives (TP) : 104
True Negatives (TN) : 57
False Positives (FP): 6
False Negatives (FN): 4

Accuracy = 94.15%
Precision = 94.55%
Recall    = 96.30%
Specificity= 90.48%

```